

# Using AI for Defense: UEBA, SOAR, Anomaly

## ML-powered defense stack

| Layer         | Tool type                 | What ML adds                                             |
|---------------|---------------------------|----------------------------------------------------------|
| NDR/NDR       | ML anomaly (Zeek → model) | New baseline behavior, not static rules                  |
| Endpoint      | UEBA / behavioral         | 'User X' profile drift → alert                           |
| SIEM/SOAR     | ML triage classifier      | Auto-prioritize, reduce noise                            |
| Vulnerability | Risk-based scoring (EPSS) | Prioritize by <i>exploit likelihood</i> , not CVSS alone |

## UEBA in practice

Build baseline: 'alice usually accesses these systems at these times'

ML flags: unusual hour, new data access, new location

Requires: good telemetry + a minimum training window (weeks)

Limitation: training-data quality = quality of detection; noisy initial FPs expected

## Anomaly-based network detection

Baseline per host: connections, bytes, timing

Flag: new destination, new timing pattern, large egress

Combine with Zeek + ML (e.g., ml-ids concepts) for beaconing detection

## Practical SOAR automation

High-confidence phishing: auto-isolate host

Low-confidence alert: auto-enrich (whois, reputation, last contacts)  
then queue for analyst

This triage automation is the *near-term AI win* for most SOC's

## Deploying ML in security (the craft)

Start with **detection-in-test** (log-only), never block

Measure F1 (precision vs recall), not just accuracy

Retrain periodically (threat drift)

Document model assumptions (the #1 audit question)

Keep a human in the loop on high-impact actions

## The honest limit

ML excels at *pattern recognition at scale*; it does not understand *intent*. The human analyst provides context; ML provides volume and speed.

AI-augmented defense means the analyst stops spending time on known-noise and starts focusing on what ML can't decide. That's the real productivity unlock.

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